

Setup

The following code sets up the system by downloading the library from the web.

Make a closed figure function

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

Make a video of drawing a face

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In [29]: # Playing with b

# Create a graph object with equal_axes)
plot = graph_funs(equal_axes=True, hide_axes=False)

# Setup the grid
# Determine the range for X based on reflections and translations.
# Determine the range for Y based on reflections and translations (look at maximum values).
plot.set_XY(minX=-10, maxX=+10, minY=0, maxY=+40)

# Draw different parts of the face and save in different frames

# Frame 1: draw the face:
make_closed_fig(plot, a=0.10, b=2, h=0, k=5, d=7, num_of_points=20)
plot.plot_funs(plot_title="Drawing the face", img_name="frame1.png")

# Frame 2: Draw the eyes:
make_closed_fig(plot, a=0.10, b=b, h=4, k=20, d=2, num_of_points=20)
make_closed_fig(plot, a=0.1, b=b, h=-4, k=20, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the eyes", img_name="frame2.png")

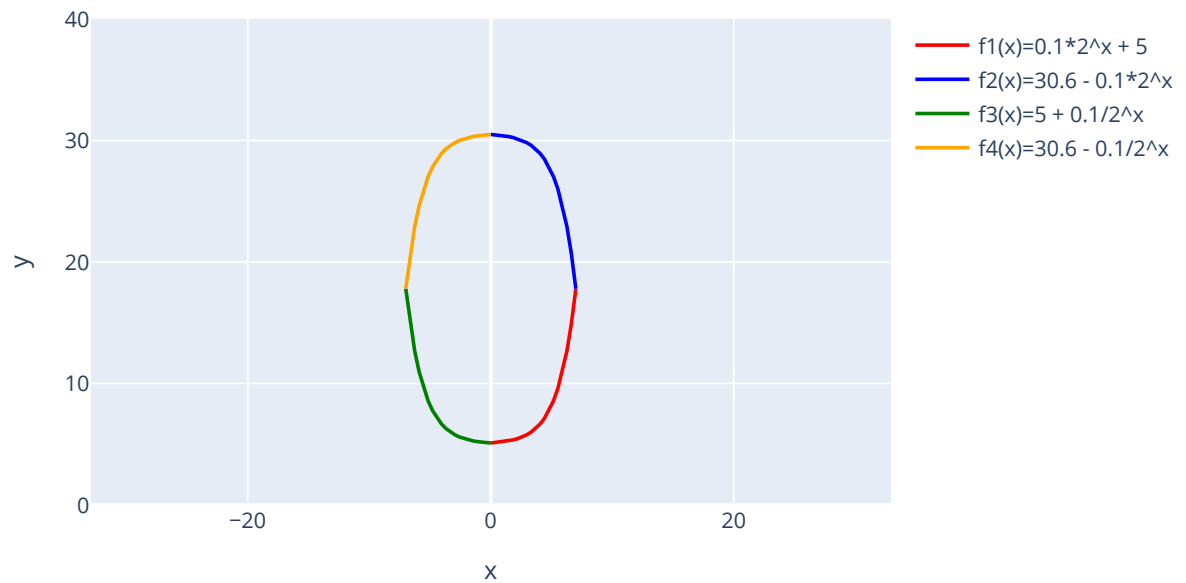
# Frame 3: Draw the nose:
make_closed_fig(plot, a=0.05, b=b, h=0, k=17, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the nose", img_name="frame3.png")

# Frame 4: Draw the mouth:
make_closed_fig(plot, a=0.07, b=b, h=0, k=5.5, d=3, num_of_points=20)
plot.plot_funs(plot_title="Drawing the mouth", img_name="frame4.png")

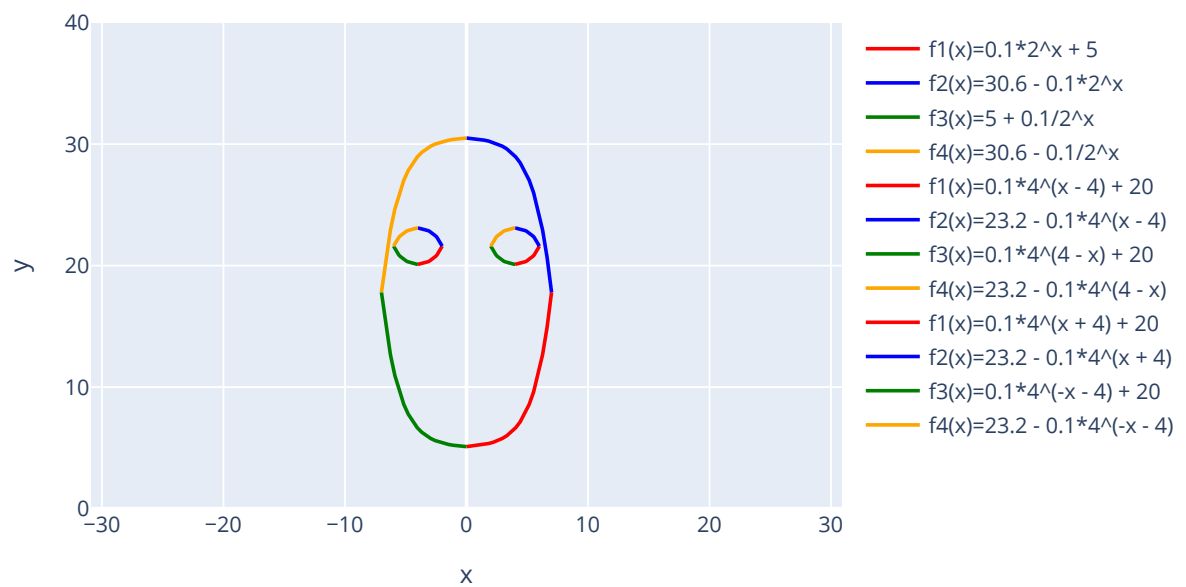
# Frame 5: Draw the mohawk:
make_closed_fig(plot, a=0.5, b=b, h=0, k=29.7, d=5, num_of_points=20)
plot.plot_funs(plot_title="Drawing the mohawk", img_name="frame5.png")

# Frame 5: Draw the makeup:
make_closed_fig(plot, a=0.25, b=b, h=4, k=18, d=2, num_of_points=20)
make_closed_fig(plot, a=0.25, b=b, h=-4, k=18, d=2, num_of_points=20)
plot.plot_funs(plot_title="Drawing the makeup", img_name="frame6.png")
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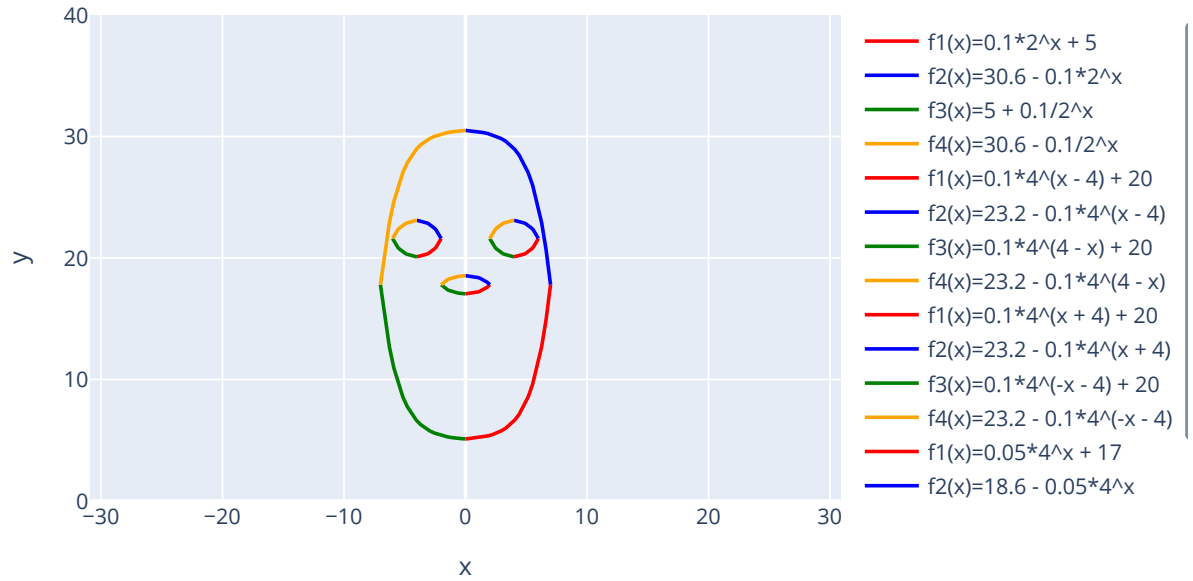
Drawing the face



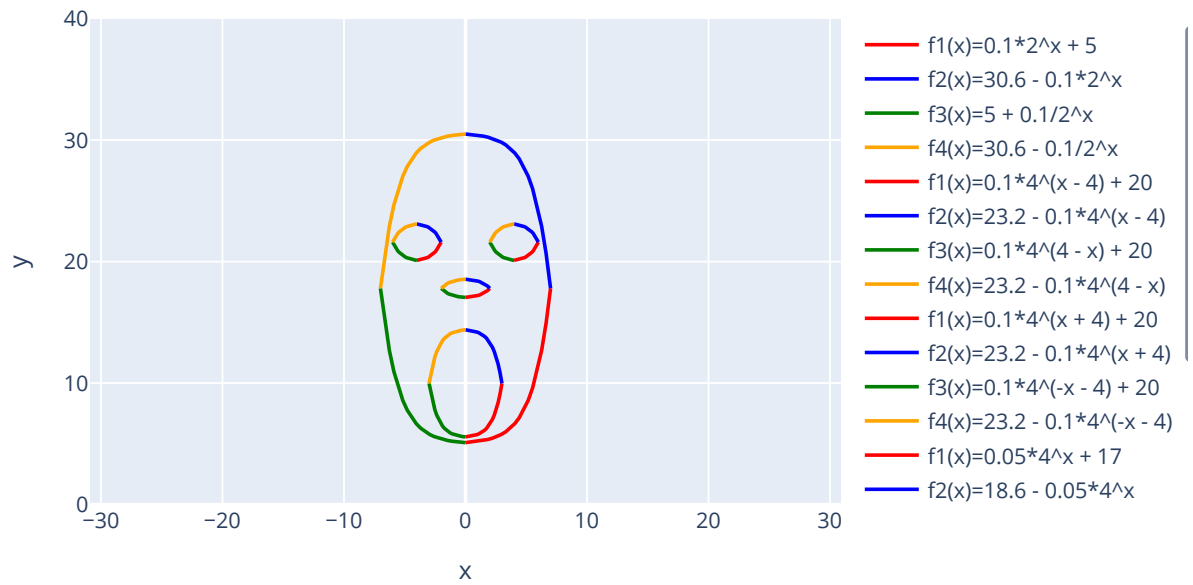
Drawing the eyes



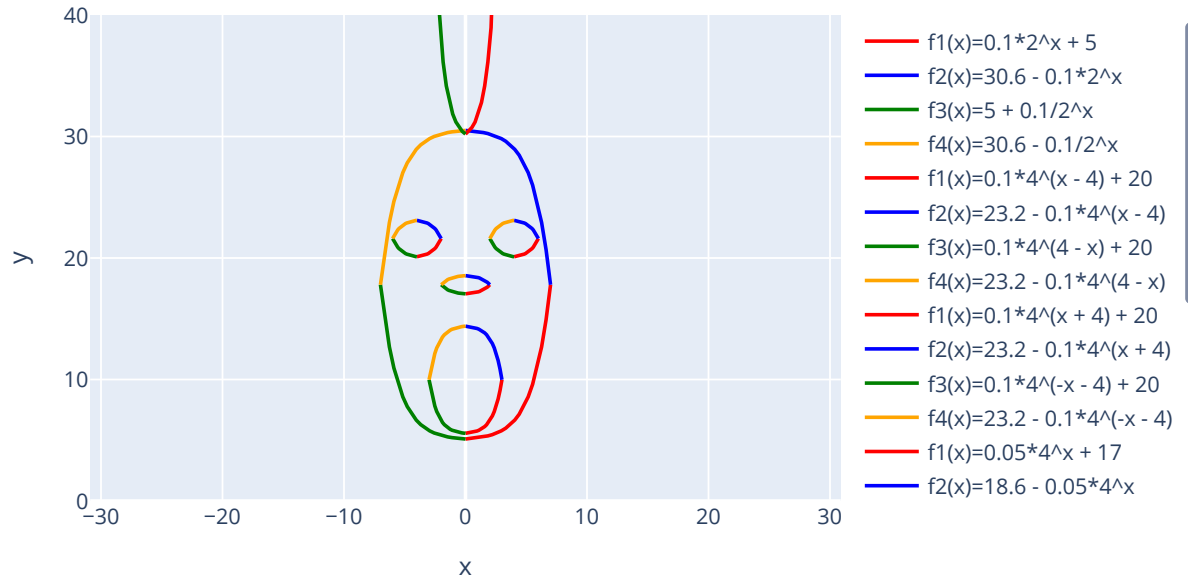
Drawing the nose



Drawing the mouth



Drawing the mohawk



Drawing the makeup

